REFA AL-AMRI

Astronomy Graduate Student at
Texas A&M University

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RESEARCH INTERESTS

Black holes, galaxy growth and evolution, black hole-host galaxy co-evolution, active galactic nuclei, dark matter.

PROGRAMMING LANGUAGES

fluent in: `Python`, TEX, Matlab.

experience with: `IDL`, `R`.

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Fall 2022 - present	**Graduate Research Assistant** Department of Physics and Astronomy	Texas A&M University
Fall 2022 - Spring 2024	**Graduate Teaching Assistant** Department of Physics and Astronomy	Texas A&M University
Fall 2021 - Spring 2022	**Undergraduate Teaching Assistant** Department of Mathematics and Statistics	Washington State University
Fall 2019	**Peer Mentor** Voiland College of Engineering and Architecture	Washington State University

EDUCATION

2022 - present	**Ph.D. Astronomy** Department of Physics and Astronomy	Texas A&M University
2022 - 2025	**M.S. Astronomy** Department of Physics and Astronomy	Texas A&M University
2019 - 2022	**B.S. Physics and Astronomy | Mathematics minor** Department of Physics and Astronomy	Washington State University

RESEARCH EXPERIENCE/ PROJECTS

Research	**Multi-Tracer Dynamical Analysis of NGC 4261 with JWST and MUSE** Using JWST NIRSpec IFU and VLT/MUSE observations of NGC 4261 to measure stellar and gas kinematics and support ongoing multi-tracer dynamical modeling of its central supermassive black hole.
Research	**JWST Studies of the Central Stellar and Gas Kinematics of M87** Using JWST NIRSpec IFU observations of M87 to measure its central stellar and gas kinematics. This work includes a first-author publication on the stellar kinematics in the central 240 pc of M87, as well as ongoing contributions to black hole mass measurements through gas- and stellar-dynamical modeling.
Project	**Modules for Experiments in Stellar Astrophysics (MESA) Exploration** Using MESA To study how rotation rate and convection mixing lengths influence the the stellar evolution of different mass stars.
Research	**Determining the Population of Blue Straggler Stars in Open Clusters** Finding blue straggler stars that are twice as massive as the main-sequence turn-off star of clusters using the GAIA database.
Project	**Exploring the Effects of Volume on Reverberation Time** Explored the effect of volume on the reverberation time using different simulated rooms while keeping the absorption coefficient and sound source constant.

OBSERVING/GENERAL EXPERIENCE

July 2025	**W. M. Keck Observatory - OSIRIS** 2025A	2 1/2 nights
December 2024	**W. M. Keck Observatory - OSIRIS** 2024B	2 1/2 nights
July 2024	**W. M. Keck Observatory - OSIRIS** 2024A	1/2 night
Feburary 2024	**W. M. Keck Observatory - OSIRIS** 2024A	1/2 night
December 2023	**W. M. Keck Observatory - OSIRIS** 2023B	2 1/2 nights
July 2023	**W. M. Keck Observatory - OSIRIS** 2023A	2 1/2 nights

TEACHING EXPERIENCE

Fall 2022 - Spring 2023	**Astronomy 101: Basic Astronomy** Department of Physics and Astronomy	Texas A&M University
Fall 2022 - Fall 2023	**Astronomy 103: Introduction to Stars and Exoplanets** Department of Physics and Astronomy	Texas A&M University
Fall 2022 - Fall 2023	**Astronomy 104: Introduction to Galaxies and Cosmology** Department of Physics and Astronomy	Texas A&M University
Fall 2022 - Spring 2023	**Astronomy 314: Survey of Astronomy** Department of Physics and Astronomy	Texas A&M University
Fall 2022	**Astronomy 401: Stars and Extrasolar Planets** Department of Physics and Astronomy	Texas A&M University
Spring 2024	**Astronomy 420: Advanced Astrophysical Research Methods** Department of Physics and Astronomy	Texas A&M University
Fall 2021 - Spring 2022	**Math 315: Differential Equations** Department of Mathematics and Statistics	Washington State University

HONORS AND AWARDS

2026	**Department Travel Award** Department of Physics and Astronomy	Texas A&M University
2022	**Department Fellowship** Department of Physics and Astronomy	Texas A&M University
2021	**Paul A. Anderson Award for Excellence** Department of Physics and Astronomy	Washington State University
2019-2022	**President's Honor Roll** Department of Physics and Astronomy	Washington State University
2019-2021	**Physics Textbook Award** Department of Physics and Astronomy	Washington State University
2019	**Boeing Peer Mentor Scholarship** Voiland College of Engineering and Architecture	Washington State University

CONFERENCES, TALKS AND PRESENTATIONS

2026	**Astronomy on Tap #85** Talk	Public Outreach Event
2026	**Astrosymposium** Talk	Texas A&M University
2026	**Black Holes and Blue Notes** Conference Contributed Talk	University of Montreal
2026	**Astronomy on Tap #80** Talk	Public Outreach Event
2025	**Astronomy on Tap #78** Talk	Public Outreach Event
2025	**Astrosymposium** Talk	Texas A&M University
2024	**Texas A&M Extragalactic Seminar** Talk	Texas A&M University
2024	**Astronomy on Tap #69** Talk	Public Outreach Event
2024	**Astrosymposium** Talk	Texas A&M University
2023	**Astronomy Bash Symposium: New Horizons in Astronomy** Poster	University of Texas at Austin
2023	**Astrosymposium** Virtual Talk	Texas A&M University
2022	**Astrosymposium** Talk	Texas A&M University
2019	**Physics and Astronomy Undergraduate Thesis Poster Session** Poster	Washington State University

PUBLICATIONS

1. *Mapping the Stellar Kinematics in the Central 240 pc of M87 with the James Webb Space Telescope.* (2026)
   Refa M. Al-Amri, Jonelle L. Walsh, Emily R. Liepold, Chung-Pei Ma, and Jenny E. Greene.
   The Astrophysical Journal, Volume 999, Number 2.
   DOI:10.3847/1538-4357/ae4233